The Instrument Is the First Suspect: Six Broken Lenses in One Empirical Program

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Abstract

Empirical claims about learned systems inherit the health of the instruments that measure them. We report six pre-publication incidents from one measurement-heavy research program on fast-weight sequence models in which a model, in one case a synthetic oracle, appeared to fail and the failure traced instead to the instrument: an admission tolerance calibrated on the wrong key population; a probe reading a layer that causal state-zeroing later proved inert; an identity-gauge assumption measured on one architecture and carried to another; an uncentered covariance estimator degenerate on the task's near-orthogonal targets; a target whose ambient identity block taxed every rank budget; and a cross-check reference invoked on the transposed side of a state convention. The same discipline adjudicated every incident, assembled across the catalogue and in full force before the final, decisive one: instrument-health adjudication before any model verdict, pre-registered crosschecks graded where the primary lens's assumptions fail, falsifier controls run to completion, and raw-artifact tiebreaks. The discipline is itself falsifiable; its decisive crosscheck survived a pre-registered shuffled-target control that would have voided the adjudication. It does not launder results: the corrected lens left the endpoint verdict standing and two genuine model failures intact. We argue the discipline transfers to any empirical evaluation gated by numerical checks, learned probes, or basis-dependent readouts.

1 Introduction

In one research program's five-day window on fast-weight sequence models, six apparent model failures traced, under the same adjudication procedure, to broken instruments rather than models (Table 1); in two further readings it confirmed the model, not the lens, had failed. An incident here is a pre-registered endpoint or evaluation round read as a model verdict; defects caught outside such a reading (the auxiliary-classifier null of §4) are noted, not counted. An evaluation pipeline that never suspects its own instruments will publish lens artifacts as model results.

Our contributions: (1) the catalogue (Table 1), each incident with its mechanism and a reusable diagnostic signature; (2) a five-rule adjudication discipline (§2) that caught all six before publication; (3) the discipline's boundary: it did not flip the pre-registered endpoint verdict, and two genuine model failures survived every instrument fix (§7).

2 The Adjudication Discipline

None of the five rules is novel in isolation; applied together, *before* any model verdict is recorded, they form a lens-failure detector with a measured record. Two clauses were codified mid-catalogue by the incidents that exposed them (Rule 3's positive-control clause, §4; Rule 2's graded crosscheck, Appendix A); all five rules were in force before the decisive adjudication of §5.

Rule 1: instrument-health adjudication precedes model verdicts. Before a sweep's pre-registered endpoint may be read, a health checklist walks the raw per-cell artifacts: gate violations, convergence, a contradiction scan between independent readouts (the rule that fired in §5).

Rule 2: pre-register a crosscheck that removes the primary lens's strongest assumption, graded wherever that assumption provably fails, the switching rule fixed in advance.

Rule 3: falsifier teeth, run to completion. Every negative control is executed, not merely written; guards are mutation-tested; every probe rung carries a planted-signal positive control; the adjudication itself carries a pre-registered falsifier whose firing voids it.

Rule 4: contradictions are resolved against raw artifacts. When two reads disagree, neither recency nor authority decides: both are recomputed from the committed raw files and the tiebreak is recorded before anything downstream runs.

Rule 5: argmax closure. Wherever a claim requires exact continuous recovery, the grading path must not be satisfiable by argmax decoding, under which a rank-1 matrix can recover on the order of d associations [Nichani et al., 2025]; the converse gap was pre-registered before it fired (§4).

3 Case I: The Tolerance That Manufactured a Cliff

A fast-weight associative memory's recall capacity is measured by exact continuous recovery (cosine ≥ 0.9, never argmax; Rule 5) across a load sweep K/d_{state}. An eval-time admission gate decides which trained cells enter a capacity fit. One leg of that gate checks that a Newton–Schulz orthogonalization in the scoring path converges to residual < 0.01 within $n_{\text{iter}} = 20$ iterations. A wide-grid sweep at $d_{\text{state}} = 96$ trained cleanly at every cell (training guard 1.0 throughout), yet only one of 12 new cells was admissible (K=78, 84, 90: all three seeds failing): the shape of a capacity cliff, contradicting the same architecture's established super-linear growth at lower state sizes [Anonymous, 2026] and the same grid's own post-fix re-fit below.

The first mechanistic diagnosis blamed the trained anchor table's convergence at high load; a follow-up refuted it with margins (residuals at most 1.6×10^{-6}, over $6{,}000\times$ below tolerance, indistinguishable across failing and passing cells; failures confined to a pool the anchor table bypasses) and the claim was formally retracted.

A purpose-built repro instrument recovered the true failing tensor: the pool's raw keys, never saved by the original checkpoint writer. Its decision rule, a pre-registered strict precedence order, was walked mechanically (Table 2): every instrument-bug branch failed to fire, and an iteration sweep resolved all 4,608 flagged episodes by $n_{\text{iter}} \leq 28$. A wall keeps failing as the budget grows; a miscalibrated tolerance resolves at a bump, and did. Root cause: a tolerance calibrated on anchor-stabilized *blended* keys, silently applied to the structurally different *raw* keys this one pool queries. Only the admission flag was recalibrated, only for the 11 cells verified to share the identical signature; the re-fit found no cliff through $K/d_{\text{state}} = 0.9375$ (per-load mean recovery 0.92 to 1.00).

4 Case II: The Instrument at the Wrong Layer

A two-block fast-weight model is trained on an associative binding task ($K = 32$ bindings per episode) against matched baselines, with two registered readouts: a discrete task metric (episode-restricted recall through the model's own LM head) and a continuous probe metric rf@0.9 (the fraction of targets a linear probe on a tapped state recovers at cosine ≥ 0.9). In the third of three consecutive rounds recorded as failures at rf@0.9 $= 0.0$, the discrete metric dissociated: recall accuracy 0.9990 against chance 0.03125 (baselines 0.0447 and 0.0295), the probe still exactly 0.0 in every arm; the dissociation, not the zero, is the finding (Rule 5).

An offline re-fit battery removed the probe-training explanations: closed-form ridge regression, pinned SGD, and MLP probes all reproduced rf@0.9 $= 0.0$ exactly (over-regularization ruled out by the rig's post-nonlinearity recovery below); the trained probe's output was the analytically known episode-membership direction (mean cosine 0.176 against the $1/\sqrt{32} \approx 0.177$ ceiling; membership cosine 0.896); the ridge fit cleared its shuffled-tap control [Hewitt and Liang, 2019] by at most $+0.059$ cosine. Meanwhile the full forward pass decodes the answer at 0.9957 accuracy ($31.9\times$ chance) for the contender, both baselines at or near chance: the answer exists in the network.

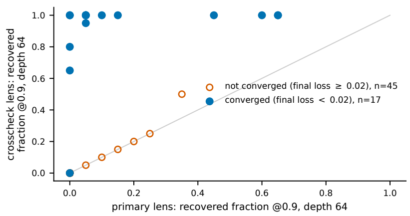

Figure 1: Primary versus crosscheck recovery (fraction recovered at cosine ≥ 0.9, depth 64), all 62 cells. Non-converged cells (open orange) sit near the origin under both lenses; every cell with disagreement ≥ 0.5 is converged (filled blue). A lens defect that activates exactly where the model succeeds is the inverse of noise.

A state-zeroing experiment [Elazar et al., 2021] located the storage (Figure 2): zeroing the first block's state S_0 collapses recall to chance (0.0286); zeroing S_1, the layer every deployed instrument tapped, leaves it bit-unchanged at 0.9990. That layer plays no causal role in the recall. Placement alone is not the whole defect: a ridge probe on the load-bearing S_0 *also* reads $\mathrm{rf@0.9} = 0.0$ (shuffled-control gaps $+0.002$ to $+0.063$), and only the post-block-1, pre-LM-head hidden exposes the recall ($\mathrm{rf@0.9} = 0.674$, gap $+0.800$), only in the arm that performs the task (ablation: 0.0; Table 3). The bindings are stored in a format no tested linear read exposes; the model's own downstream nonlinearity is the only decoder found.

The same diagnosis exposed a second defect in passing: an auxiliary classifier whose labels were uniform given the decoded quantity, so a perfect tap scores chance; its expected-null had been read as confirmation for a round, hence Rule 3's positive-control clause.

5 Case III: The Gauge That Did Not Transfer

A recurrent composer is trained on the word problems of five finite groups and scored for compositional depth generalization: trained at depth ≤ 8, tested at depth 64. Two pre-registered lenses score recovery of the group representation from the state: the *primary* de-gauges with a scale-only correction, valid when the learned basis matches the target up to scale ($\hat{Q} \approx I$, measured on every checkpoint of the *previous* architecture); the *crosscheck* fits a full orthogonal intertwiner by Procrustes, fit and scored on disjoint splits, assuming nothing about the gauge.

The signature. The 62-cell sweep completed cleanly; run as built on the primary lens, the endpoint machinery returned FALSIFY: no measurable separation from the baseline at the decisive groups. The mandatory instrument-health pass (Rule 1) surfaced a contradiction (Figure 1): wherever training had *not* converged, both lenses read near zero; wherever it converged, they contradicted each other at full magnitude (one converged cell: loss 0.0001, primary 0.050, crosscheck 1.000). All 13 cells with a lens disagreement of at least 0.5 are converged cells (threshold-robust: every cell with a disagreement above 0.3 is converged, 16 of 16); on all 25 Arm-2 baseline cells (5 groups $\times$ 5 seeds, 50 checkpoint values) the lenses agree exactly. The harvest stopped for adjudication with no model verdict recorded.

The tiebreak. A leaky crosscheck reading high on junk would leave the falsification standing. The adjudication, against raw artifacts and code (Rule 4), rested on four grounds: (1) the crosscheck is leakage-guarded by construction (fit and eval on disjoint splits); (2) it discriminates, reading near zero on every non-converged cell; (3) the primary's scale-only correction was already flagged as basis-brittle across seeds (per-seed mean cosine 0.03 to 0.69 where the crosscheck read 0.86 to 0.95) and its $\hat{Q} \approx I$ evidence came from a different architecture (a perfect model in a rotated basis reads zero under an identity gauge); (4) the synthetic acceptance test already grades the crosscheck when the injected gauge is non-identity (Rule 2). Verdict: the primary lens is broken on converged cells; the mechanical falsification is void *as a model verdict*.

Teeth. The tiebreak carried its own pre-registered falsifier (Rule 3): shuffled-target re-scoring must read near zero, or the tiebreak is wrong and the contradiction escalates to an instrument rebuild. On three pre-registered converged checkpoints spanning three groups (among them the converged exemplar above), the shuffled crosscheck read 0.00, 0.00, and 0.05, far below the 0.5 void threshold, while the unshuffled reads reproduced the committed values bit-identically (Table 5).

The re-metric. The full grid was re-read under the crosscheck lens through the unmodified verdict machinery, which first reproduced the committed primary verdict and all 62 recomputed ceilings to within 10^{-9}. The corrected endpoint keeps the same FALSIFY verdict *and the same machine-*

emitted reason (§7, Table 4); what the fix changes is the readable geometry: the primary was not a blanket null (S3 cleared its bar, 0.54 against 0.459) but masked which groups fail, while under the crosscheck the two decisive failures, an S5 near-miss and an A6 non-convergence, stand out lens-independently. The verdict itself never moved; fixing the instrument changed how to read it.

6 Three More Lenses, Briefly

Three further incidents (rows 4–6 of Table 1; numbers in Appendix A): an **uncentered covariance**, degenerate on the task's own near-orthogonal targets, that failed a synthetic *perfect* model (the rescue de-confounds into centering plus a full-intertwiner degauge); a **target with an ambient identity block**, under which a causal rank experiment returned near-chance recovery at every rank (zero in four of five groups; 0.17 and 0.08 in the fifth) while the capped arms sat at the exact rank-k ceiling $\sqrt{k/d_{\mathrm{state}}}$: undelivered by the instrument, not failed by the model; and a **reference kernel invoked on the transposed side** of a state convention, "contradicting" a correct recurrence at the $\sqrt{2}$ signature of a transposed outer product, exonerated by a closed-form hand computation.

7 The Boundary: What the Discipline Did Not Flip

A lens-repair procedure that only rescued the preferred hypothesis would be a laundering machine; under the corrected lens, Case III's endpoint remained FALSIFY, held there by two lens-independent model results: one decisive configuration never reaches training-support convergence (recovery 0.00 at depth 64, all five seeds, both lenses), and the other decisive group sits below its pre-registered bar (0.483 against 0.735) lens-independently: one seed generalizes at 0.00 under both lenses despite low training loss, and even its siblings reach only 80% and 65% of their own ceilings, under the 90% criterion, so the group misses its bar with or without the outlier, against a baseline pinned at 0.0.

Case I shows the same refusal: a diagnosed pool-margin confound ($+0.0330$, $12\times$ the 0.0028 noise floor) was withheld because the diagnostic's own cells never cleared admission, and an archived exact-ceiling reading (1.0000, three seeds) that did not replicate at a fresh seed (0.9725) was re-scoped to the seeds measured; the machinery that unlocked eleven quarantined cells declined, twice, to promote findings it could not support.

8 Related Work and Limitations

Related work. Silberzahn et al. [2018] and D'Amour et al. [2022] document divergent conclusions from underspecified pipelines; here the models are fixed and the *measurement pipeline* is the adjudicated object. Henderson et al. [2018] and Bouthillier et al. [2021] quantify seed and tuning variance; none of the six defects is sampling variance, so variance accounting catches none. REFORMS [Kapoor et al., 2023], Sculley et al. [2018], and Lipton and Steinhardt [2018] prescribe what a trustworthy paper discloses; we supply the in-flight machinery such checklists presume exists. The probing literature supplies the instrument-hygiene checks the wrong-layer case rediscovers at evaluation scale: control tasks and selectivity [Hewitt and Liang, 2019], the probe-versus-behavior dissociation [Belinkov, 2022], amnesic probing [Elazar et al., 2021], and randomization-based sanity checks [Adebayo et al., 2018]. Schaeffer et al. [2023] argue the inverse of our thesis (an apparent model *property*, emergence, as metric artifact) from the same instrument-first stance; each diagnoses one instrument class, where this catalogue is a serial, pre-registered, cross-instrument adjudication record with falsifier teeth. Nichani et al. [2025] prove Rule 5's argmax/exact-recovery gap; we report it firing live (§4); the fast-weight testbeds follow Schlag et al. [2021].

Limitations. All six incidents come from one program, one team, one architecture family: they support no prevalence claim, and we cannot report the rate a lighter process would miss. The catalogue is survivor-biased: a defect missed by both crosscheck and falsifier battery appears in no record we could write. It is biased by trigger too: the discipline fires on apparent model failure, so defects flattering the model are under-represented; five of six fixes moved results in the program-favorable direction. In the one incident costed, the two diagnostic instruments took 0.50 and 1.10 GPU-hours (archived wall-clock) against the 6.33 GPU-hours of training the uncorrected readings would have discredited. We report no false-alarm rate: the records do not tally escalations against clean instrument-health passes, so the precision of "suspect the instrument first" is unmeasured.

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A The Three Brief Lenses, in Full

The covariance that could not see. An acceptance test for a representation-recovery pipeline injected a synthetic *perfect* model and failed it. The subspace step estimated an uncentered covariance, degenerate on the task's own targets: for near-orthogonal states $Z \approx c(\rho \oplus I)$, $ZZ^\top \approx c^2 I$ is isotropic and carries no subspace information. On the production harness, the headline rescue of the oracle injection (mean cosine 0.084 to 0.9996) folds in two registered lens changes: centering *and* a switch from the scale-only primary to the full-intertwiner crosscheck. Isolating centering under a matched full-intertwiner metric moves mean cosine 0.705 to 0.9996; under the scale-only lens the centered pipeline reads 0.046, no better than the uncentered 0.084, because the injection's true gauge is non-identity, the pre-echo of §5. Centering is a genuine fix on its own, but the full rescue needed a second step on top of it: a full-intertwiner degauge. The fix (centering) cancels the constant identity block because the group mean of a nontrivial representation vanishes; the acceptance injection was extended so the previously skipped subspace step now sits inside the test.

The target that taxed every budget. A causal test of a rank-recruitment law capped model rank at k around the predicted minimum $d_{\min}$ and expected recovery to return at $k = d_{\min}$. Every capped arm returned near-chance recovery on both lenses: zero in four of the five groups, and 0.17/0.08 in the fifth (S3), far below that group's ceiling. The target had been constructed as $\rho \oplus I$: the ambient identity block inflates the as-built target rank by two, so every rank budget buys the constant block first. The diagnosis is quantitative: a rank-k model's best achievable cosine against this target is exactly $\sqrt{k/d_{\text{state}}}$, and 37 of 39 force-rank cells sat within 0.07 of that ceiling (mean absolute deviation 0.028); the arms had trained to their theoretical optimum against the wrong target. The causal test was undelivered by the instrument, not failed by the model; a corrected-target, zero-padded re-run subsequently delivered the test (companion work, anonymized) [Anonymous, 2026].

The reference on the wrong side. A bespoke recurrence was cross-checked against a reference kernel and disagreed at relative Frobenius residual 1.40; the recurrence was the initial suspect. At depth one the update is closed-form ($S_1 = \beta v k^\top$), so a hand computation with scalar loops adjudicates conventions with no numerics: the composer matched the closed form at 4.5×10^{-8} and its *transpose* at 1.405, and for near-orthogonal unit k the transpose signature is exactly $\|vk^\top - kv^\top\|_F / \|vk^\top\|_F \approx \sqrt{2}$. The reference library's documented state convention is the transpose of the pinned equation; the cross-check's own reference invocation was the defect. With the returned state transposed before comparison, the cross-check passes all three pinned configurations (residuals 2.8×10^{-3} to 4.5×10^{-3}), and a permanent negative test confirms a transposed-update mutant is killed at 1.405.

B The Catalogue and Per-Incident Detail Tables

Table 1 is the six-incident catalogue in one view; the remaining tables and Figure 2 carry the per-incident numbers the body summarizes. Every value traces to the same evidence rows (and raw artifacts, with checksums) recorded in the paper's claims-to-evidence map.

Apparent model failure	Actual instrument defect	Diagnostic signature	Resolution
Capacity collapse, 11/12 cells quarantined (§3)	tolerance calibrated on blended keys, applied to raw keys	every flagged episode resolves under an iteration bump	flag recalibrated; no cliff
Three failed evaluation rounds (§4)	probe tap on a causally inert layer; storage not linearly decodable	task metric near ceiling, probe zero; state-zeroing localization	metric re-registered; tap relocated
Generalization endpoint falsified (§5)	identity gauge from one architecture carried to another	primary/crosscheck contradiction, only on converged cells	crosscheck decisional; verdict re-read (still falsified)
Perfect synthetic model fails acceptance (§6)	uncentered covariance, degenerate on near-orthogonal targets	oracle injection scores near zero; centering restores it	estimator centered; injection extended
Causal rank test near-chance at every rank (§6)	target built with an ambient identity block, taxing every rank budget	capped cells sit at the exact ceiling $\sqrt{k/d_{\mathrm{state}}}$	target zero-padded; test rerun
Recurrence contradicted by reference kernel (§6)	reference invoked on the transposed side of a state convention	relative residual $\approx \sqrt{2}$; closed-form hand computation	reference transposed; mutant-kill check added

Table 1: The six incidents: in each, the model was initially the suspect and the instrument was the finding, and every row was adjudicated against raw artifacts before any model-level conclusion was recorded. Sections named per row.

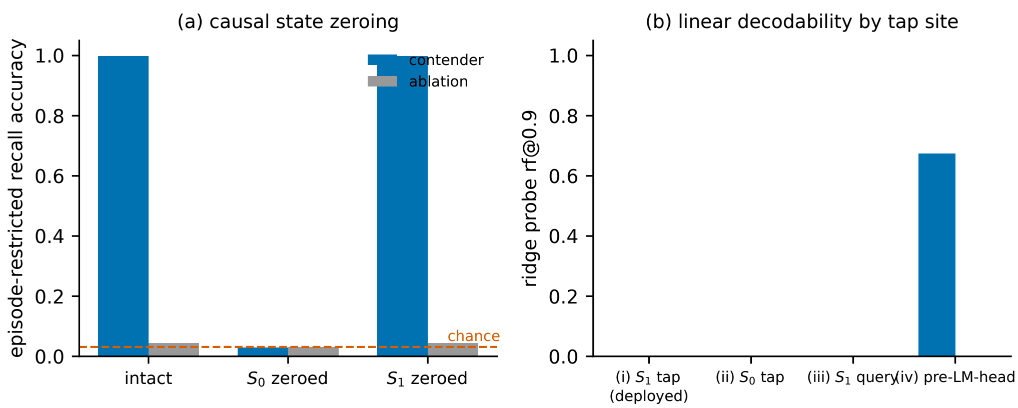

Figure 2: Case II, the wrong-layer instrument adjudicated causally. (a) Episode-restricted recall accuracy under state zeroing, contender (blue) and ablation baseline (gray), chance $= 1/32$ dashed: zeroing S_0 collapses recall to chance in both arms; zeroing S_1, the layer every deployed probe tapped, changes nothing. (b) Linear (ridge) probe recovery rf@0.9 by tap site: both recurrent states read zero in both arms, including the causally load-bearing S_0; only the post-nonlinearity pre-LM-head hidden exposes the recall, and only in the arm that can perform the task. The figure regenerates from the archived localization summary under an asserted checksum.

Step (pre-registered order)	Outcome	Reading	Rules out
entity-set reconstruction gate	pass	$107/107$ set-equal	wrong-partition analysis
pool-membership precheck	pass	0 violations / 36 events	instrument bug (structural)
reproduction-count floor	pass	$12\times$ the minimum	fluke event
live/offline agreement	pass	0 disagreements	instrument bug (numerical)
iteration sweep	resolves	4,313 @ n_{iter}=24; 295 @ n_{iter}=28	structural wall

Table 2: Case I, the repro instrument's decision-rule walk on the real cell (K=84): each step ran in its pre-registered precedence order, and the terminal verdict (tolerance miscalibration) was reachable only after every instrument-bug branch failed to fire. All 4,608 flagged episodes resolve by $n_{\mathrm{iter}} = 28$; none remained at larger budgets. Evidence row I3.

tap site	contender		ablation	
	rf@0.9	gap	rf@0.9	gap
(i) S_1, shallow query (deployed)	0.0	$+0.060$	0.0	$+0.005$
(ii) S_0, bind pathway	0.0	$+0.006$	0.0	$+0.002$
(iii) S_1, true query pathway	0.0	$+0.063$	0.0	$+0.005$
(iv) pre-LM-head hidden	$\mathbf{0.674}$	$+\mathbf{0.800}$	0.0	$+0.006$

Table 3: Case II, ridge-probe recovery by tap site; "gap" is each fit's cosine-mean margin over its own shuffled-tap control. No state-level linear tap recovers anything in either arm, including (ii) on the causally load-bearing state; the post-nonlinearity tap (iv) recovers strongly and only in the arm that performs the task. Evidence row W4.

group	ceiling (X)	far (X)	bar	clears	baseline far (X)	separates
S3	0.920	0.900	0.828	yes	0.150	no
S4	1.000	1.000	0.900	yes	0.000	yes
A5	0.490	0.390	0.441	no	0.000	no
S5 (decisive)	0.817	0.483	0.735	no	0.000	no
A6 (decisive)	0.190	0.000	0.171	no	0.000	no

Table 4: Case III, the re-metric: the pre-registered endpoint computed under the crosscheck lens (X) per group, by the unmodified verdict machinery on shadow cells; "far" is depth 64 and "bar" is 90% of each group's own ceiling. The endpoint remains FALSIFY (neither decisive group separates), but the failure geometry is now readable: A6 never converges at its decisive configuration under either lens, and S5 misses its bar lens-independently (one seed generalizes at 0.00; its two siblings reach only 80% and 65% of their own ceilings, both below the 90% bar). Separation requires the contender to clear its bar *and* the Arm-2 baseline to collapse below 50% of its own ceiling; S3 clears its bar but its baseline does not collapse ($0.15 > 0.125$), hence no separation. Evidence row X5.

converged control checkpoint	real	shuffled	falsifier (≥ 0.5) fires
A6, n_h=4, seed 0	1.000 (bit-identical)	0.000	no
S5, n_h=4, seed 0	0.800 (bit-identical)	0.000	no
S4, n_h=2, seed 2	1.000 (bit-identical)	0.050	no

Table 5: Case III, the tiebreak's own pre-registered falsifier: on three converged checkpoints, the crosscheck re-scored with targets shuffled across the item pool reads far below the 0.5 void threshold, while the unshuffled reads reproduce the committed values exactly ("bit-identical"). Had any shuffled read cleared 0.5, the tiebreak was pre-registered to be wrong and the adjudication void. Evidence row X4.